P · ARCTICPHONE

AF-P-101

CONCEPT — RURAL COMMS DEVICE

REV 0 · 2026-11-12

MESH-NATIVE PHONE FOR RURAL & REMOTE CANADA

CONCEPT — ARCHITECTURE NOT FROZEN

WHAT CHANGED VS MONOLITH

- Monolith was a flagship research device — titanium spine, optical docks. Wrong phone for a rancher in a -35 °C truck cab.
- ArcticPhone is designed rural-first: mesh-native, satellite SOS-ready, offline-first, repairable, community-owned.
- Every ArcticPhone IS a node — calls, texts, data ride the community mesh when there's no cell signal.
- The phone respects that rural life is often offline life — it works when everything else has failed.

ENVELOPE TARGET

DISPLAY
glove-friendly UI

~145×72×14 mm · ~220 g

ruggedized · replaceable battery

ARCHITECTURE OPTIONS (DECISION REQUIRED)

Opt	Approach	Pros	Cons
[illegible]	[illegible] mesh HAT (SX1262+MCU)		
[illegible]	[illegible] SX1262)		
[illegible]	[illegible]-compatible layer		

CAPABILITY SET

Capability	Implementation
[illegible]	[illegible] LoRa 915 + signed envelopes — every phone extends the network
[illegible]	[illegible] messenger module (inReach-class now; own sats later) — SOS where nothing works
[illegible]	[illegible]s, forms, messaging EN/FR — useful with zero connectivity
Cellular	LTE Cat-1/4 when towers exist — off-the-shelf modem path proven on L1
Power	Replaceable battery + USB-C PD + solar trickle option
Rugged	IP68, -30 °C operating, glove UI, MIL-STD-810H drop test target

OPEN QUESTIONS

- Architecture A/B/C decision — recommend A for speed, B for sovereignty.
- Satellite SOS: buy module now (inReach-class) vs wait for own constellation.
- Display: transflective sunlight-readable vs standard LCD + high-nit.
- Regulatory: phone certification (ISED) is a real cost — budget it.

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SCALE NTS · SHEET 1/2

DESIGNED FOR RURAL & REMOTE CANADA — -45..+70 °C · IP65 · SOLAR-FIRST · COMMUNITY-OWNED

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CONCEPT — SPECIFICATION SKETCH

TARGET SPEC + BOM CATEGORIES + ROADMAP

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CONCEPT — ARCHITECTURE NOT FROZEN

TARGET SPECIFICATION

Parameter	Target	Parameter	Target
[illegible] 14 mm			
[illegible] GNSS			
[illegible] (+ eMCU)			
[illegible] 4000 mAh			
[illegible] Linux			
[illegible] 55°C			
[illegible] back			

BOM CATEGORIES (NOT FIXED — ARCHITECTURE PENDING)

Category	Options / notes
Compute + display	Rugged Android module (option A) OR custom board (option B)
Mesh module	SX1262 + nRF52/ESP32 coprocessor — same envelope as fleet
Satellite	Garmin inReach module class, OR Iridium Certus OEM, OR future own-sat
Cellular	LTE Cat-4 modem + SIM slot — proven path from L1 HAT
Power	4000 mAh LiFePO4 pouch, USB-C PD trigger, solar pad option
Enclosure	IP65 + TPU bumpers, gasketed ports, captive screws

ROADMAP TO FIRST UNIT

- Phase 0 (now): architecture decision + industrial design concept — this sheet.
- Phase 1: prototype on dev-kit + mesh HAT — prove mesh+SOS on existing hardware.
- Phase 2: custom enclosure + integration — 3D print → machined → mold decision.
- Phase 3: certification (ISED/RF) + pilot batch — 50–100 units for founding communities.
- Phase 4: volume — repairable design pays off at scale (modules, not glue).

UNIQUE vs EVERY OTHER PHONE

- The only phone that is a FIRST-CLASS MESH CITIZEN — your phone extends the network just by existing.
- Satellite SOS as a core feature, not an accessory — designed for the places towers aren't.
- Offline-first: apps, maps, forms, messaging all useful with zero bars.
- Community-owned: the network your phone joins belongs to your community, not a corporation.
- Repairable by design — a local tech fixes it with hand tools.

RELATIONSHIP TO OTHER NODES

- Phone = personal node; L1 = field/shared node; Totem = public node; Mite = environmental; Anchor = infrastructure.
- All speak the same signed mesh — a call can route phone→phone→brick→satellite.
- Phone is the highest-volume node — its mesh radio is the network's user terminal.

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